

Magnetism

Ferrous materials: Metals that can be attracted to magnets, and can be used to make magnets.

Example: Nickel, cobalt, iron, steel.

Soft metals:

Metals that can be magnetised and de-magnetised easily.

→ Iron

Uses: * Electromagnet
* Transformer

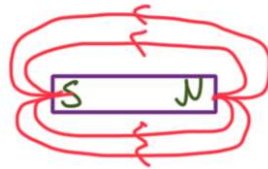
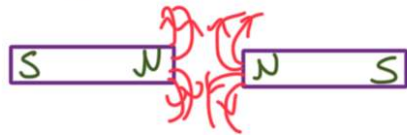
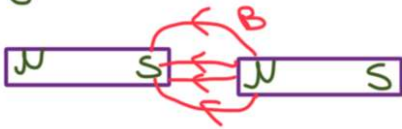
Hard metals:

Metals that can be magnetised and de-magnetised hardly / difficult

→ Steel

Uses: * Permanent magnet
* Generators
* Motors

Magnetic Field: A path of a north pole if placed in a field.



How to know the shape of the magnetic field?
Use iron filings around the magnet.

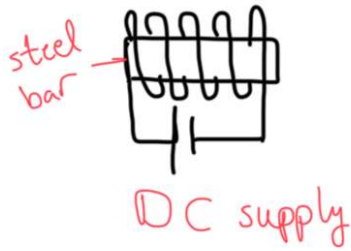
How to know the direction of the magnetic field?

Use a compass, which shows the direction of the magnetic field.

How to magnetise a metal

① Swiping: 

② Solenoid: Applying a large amount of direct current (D.C) for a long time, then either:



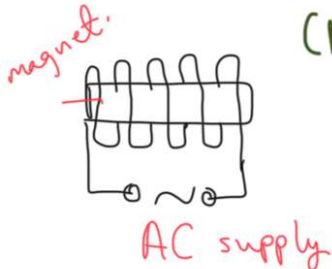
- 1- Remove the bar out slowly
- 2- Reduce the current slowly to zero

How to de-magnetise a magnet?

1- Hammering

2- Heating.

3- Solenoid: Applying a large amount of alternating current (AC) supply, then either:



- 1- Remove the bar slowly
- 2- Reduce the current slowly to zero

Magnetic Field produced by a current:



Into the
page

-Z



Out of
the page

+Z

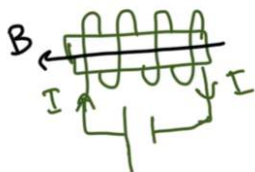
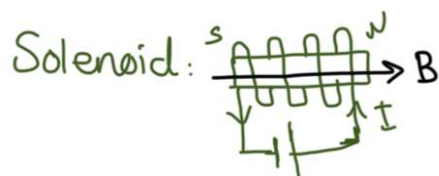
To know the direction of the magnetic field (B), we use Fleming's right hand rule (like rule). The thumb shows the current, so the fingers rotate with the magnetic field

→ To increase the magnetic field:

1- Increase the current

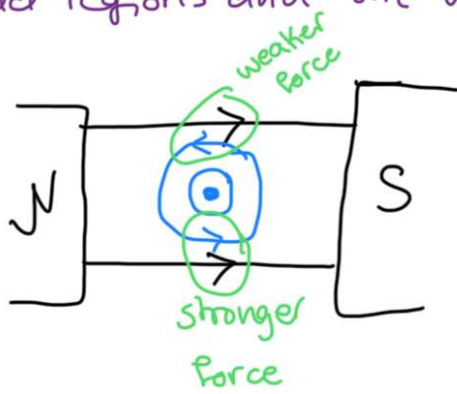
2- Reduce the radius

3- More coils

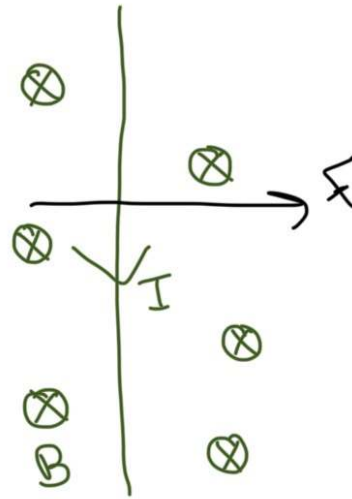
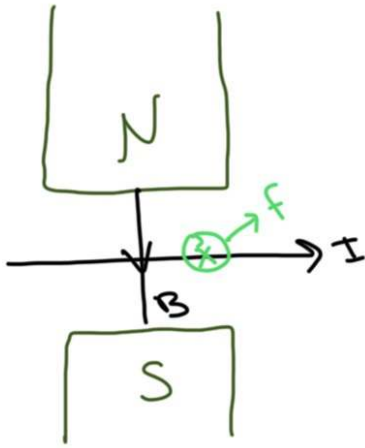


Magnetic force = Current in + Magnetic field
a wire

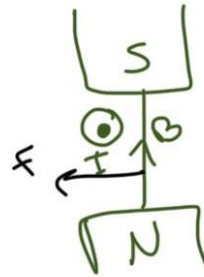
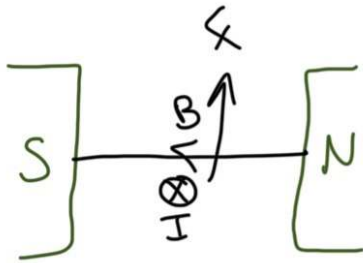
If a wire is placed in a magnetic field, the interaction between the two fields produce a stronger and a weaker magnetic field regions and the wire moves (force) from high to low.



The direction of the magnetic field force is known by Fleming's left hand rule. (FBI)



- Direction of the force is always opposite to the motion of the wire



- 1- Flipping the current or the field $\rightarrow$ Reverses the force
- 2- Flipping both the current and the field $\rightarrow$ keeps the force same.

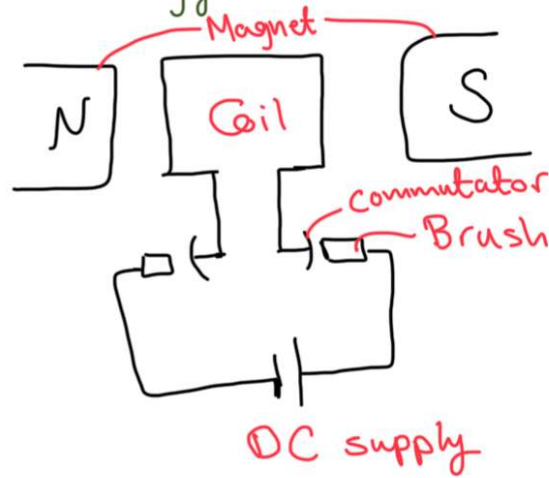
* To increase the force:

- 1- Increase the current
- 2- Use stronger magnetic field.

DC motor: Converts electric energy into the kinetic energy.

Consists of:

- 1- Magnet
- 2- Coil
- 3- Commutator
- 4- Brusher
- 5- DC supply



Explain: The interaction between the current in the coil and the magnetic field.

- * The interaction produces two equal and opposite magnetic forces, but not in the same line of action, so the coil rotates.
- * The job of the commutator is to reverse the current each half rotation, to keep the rotation in the same direction.
- * The job of the brushes is to conduct current to the coil.
- * To rotate faster: Increase the current, or the number of coils, or use a stronger magnet.

To reverse the direction of the rotation:

- 1- Reverse the battery
- 2- Flip the magnet.

→ The minimum moment is when the coil is perpendicular to the magnetic field.

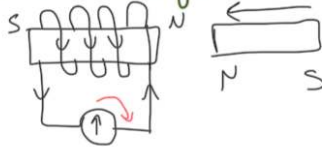
→ The maximum moment is when the coil is parallel to the magnetic field.

Electromagnetic induction: The production of induced current / emf by changing the magnetic field or by cutting the magnetic field lines by a wire.

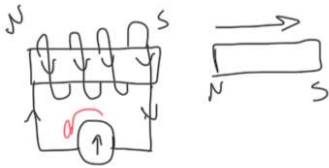
To increase the induced current:

- 1- Increase the speed of the change
- 2- Stronger magnet.
- 3- More coils.

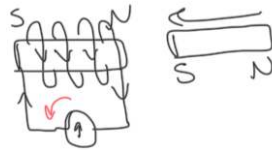
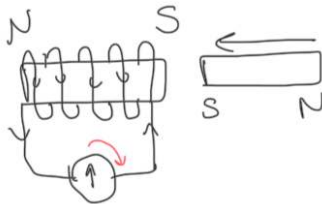
Lenz law: The induced current will be in a way to oppose the change.



* If the N-pole is moving towards the coil, an induced current is created to make an induced N-pole to oppose the change by repulsion so the voltmeter deflects to the right.



* If the N-pole is moving away from the coil, the induced current is created to make an induced S-pole to oppose the attraction, so the voltmeter is deflected to the left.

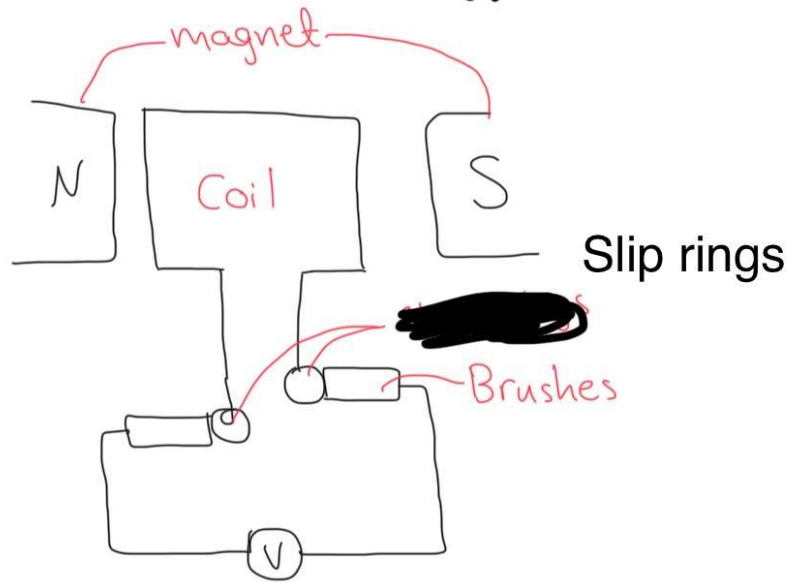


In case of a straight wire cutting the magnetic field lines, the current is induced. So right hand rule

AC generator: It converts the kinetic energy into electrical energy.

It consists of:

- 1- Coil
- 2- Magnet
- 3- Brushes
- 4- ~~Slip rings~~
Slip rings



Explanation: The rotation of the coil inside the magnetic field cuts the magnetic field lines, which produces an induced current/emf.

*The current changes its direction each half rotation, so it's called alternating current. (AC)

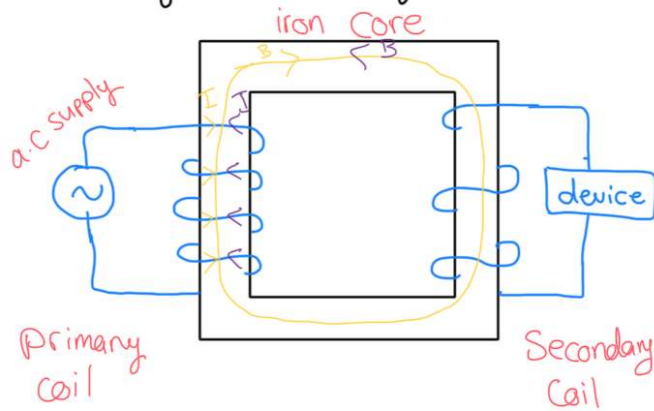
*The current/emf is max when the coil is parallel to the magnetic field

*The minimum current/emf (zero) is when the coil is perpendicular to the field lines.

To create a larger current/emf:

- 1- Stronger magnet
- 2- More coils
- 3- Faster rotations.

Transformers: Devices that use electromagnetic induction to change the voltage

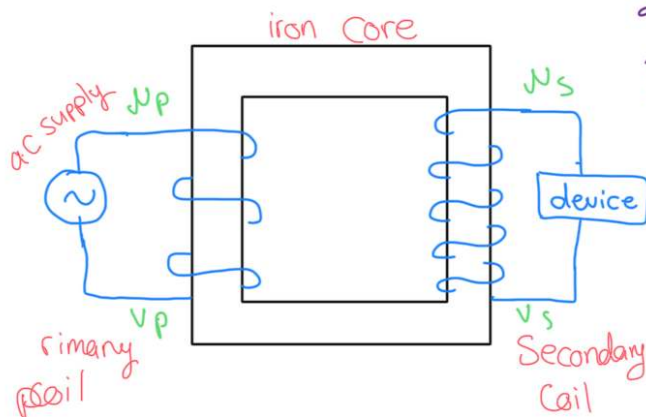


Explanation: AC supply provides alternating current which produces a changing magnetic field that passes through the iron core to pass through the secondary coil. Since there is a changing magnetic field through the secondary coil, induced current I_s is produced.

Two types of transformer:

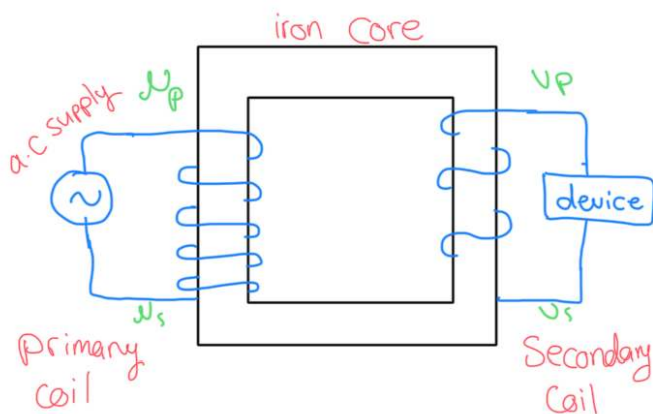
1- Step-up transformer: where:

- 1- $N_p < N_s$
- 2- $V_p < V_s$
- 3- $I_p > I_s$



2- Step-down transformer: where:

- 1- $N_p > N_s$
- 2- $V_p > V_s$
- 3- $I_p < I_s$



Transformer equation:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

V: Voltage

N: Number of coil

→ In transformers, the current is inversely proportional to the voltage. as the power is constant in: $I = \frac{P}{V}$

→ Power in primary coil = Power in secondary coil

$$P_p = P_s \rightarrow I_p V_p = I_s V_s$$

National Grid:

Power station $\xrightarrow{25KV}$ Step-up $\xrightarrow{400KV}$ Step-down $\xrightarrow{220V}$ Home
6MW transformer transformer

The use of the step-up transformer to increase the voltage to:

1- Reduce the current in the wires so the power that is lost is much less.

2- The reduction in the current means we use a thinner wire, so less cost